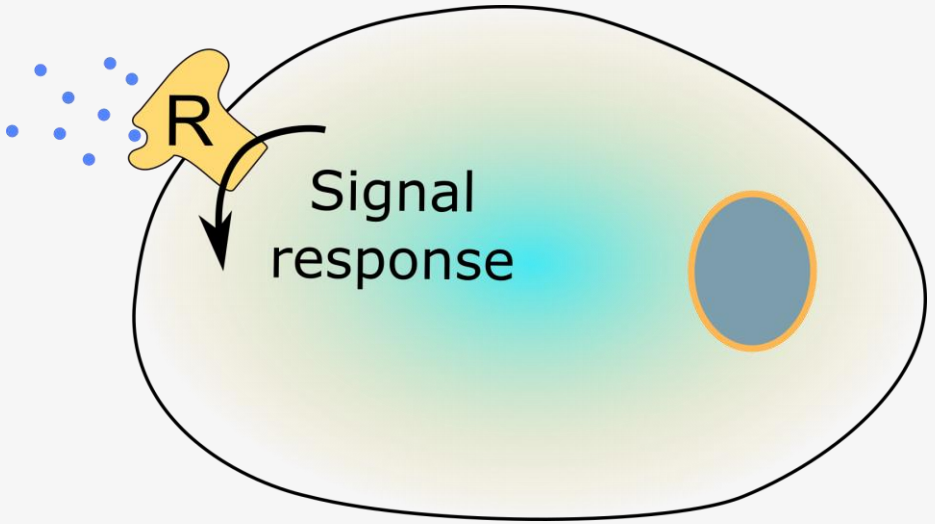
PHAR0002: More Drug Targets

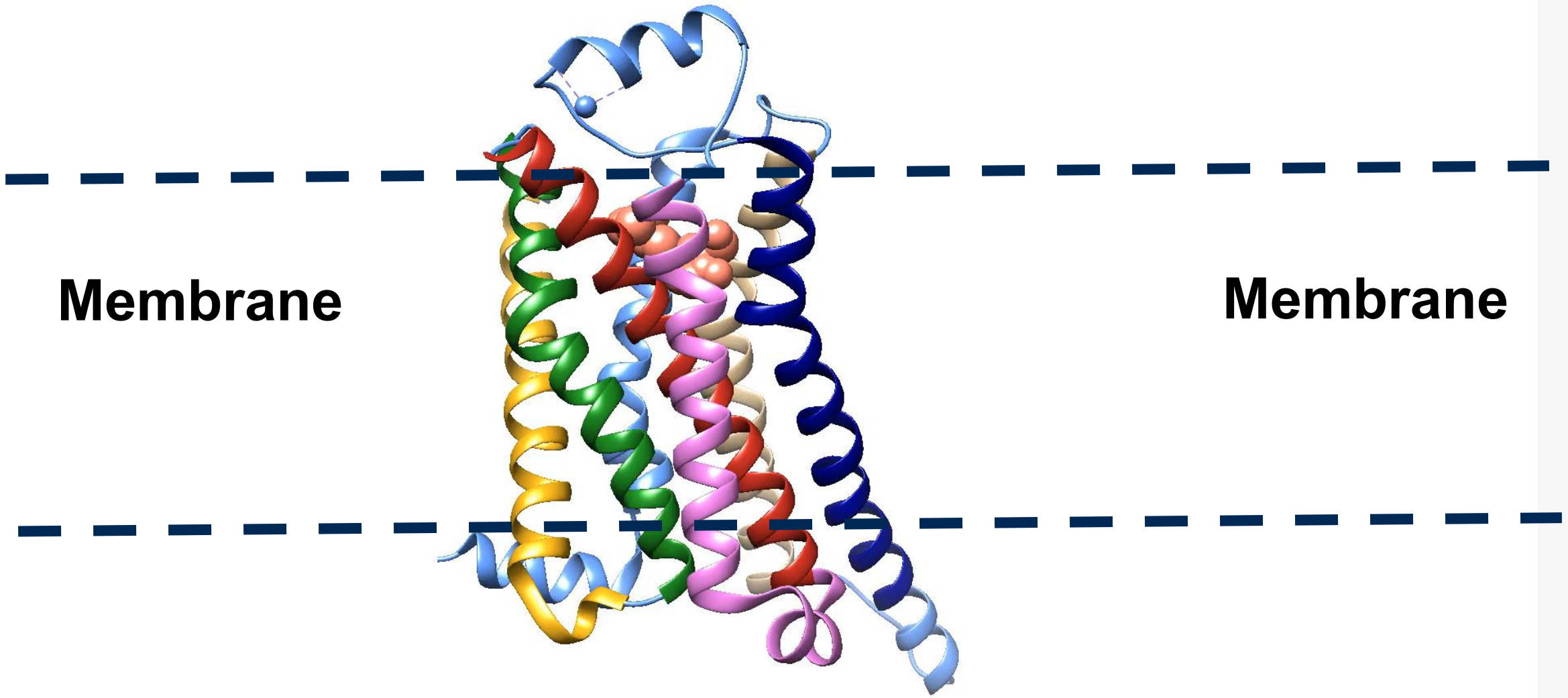
Dr. Guy Moss

1. G Protein-Coupled Receptors

2. Receptor Tyrosine Kinases



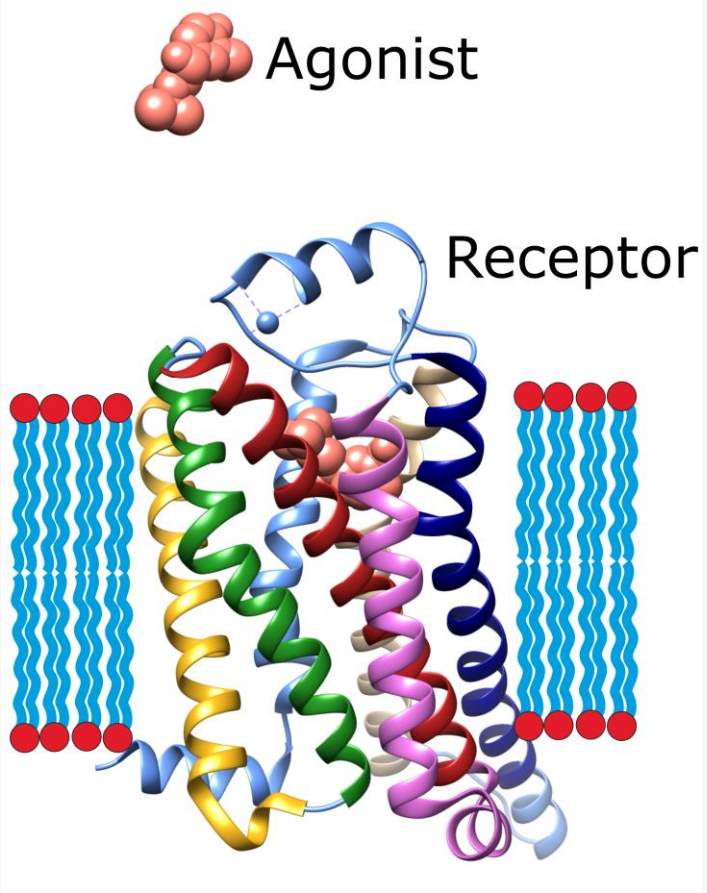
G Protein-Coupled Receptors.



GPCRs or 7TM receptors because they have seven transmembrane helices...

G-Protein Coupled Receptors.

A huge protein family (>900 in the human genome).



GPCRs as pharmacological targets..

Sumatriptan (commercial name Imigran) for the treatment of migraine.

Cetirizine (commercial name Zirtek) a commonly used antihistamine.

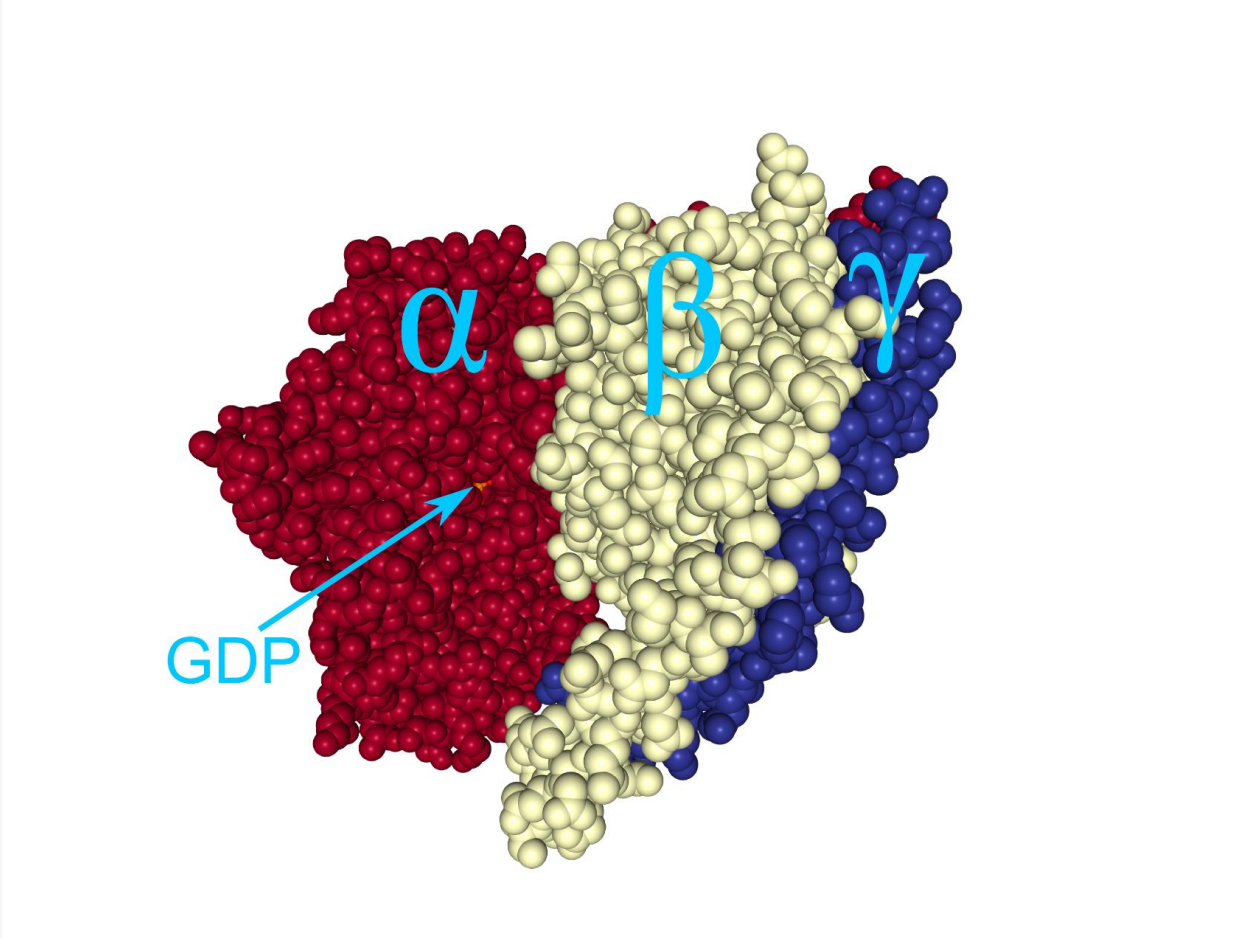
Salbutamol (commercial name Ventolin) for the treatment of asthma.

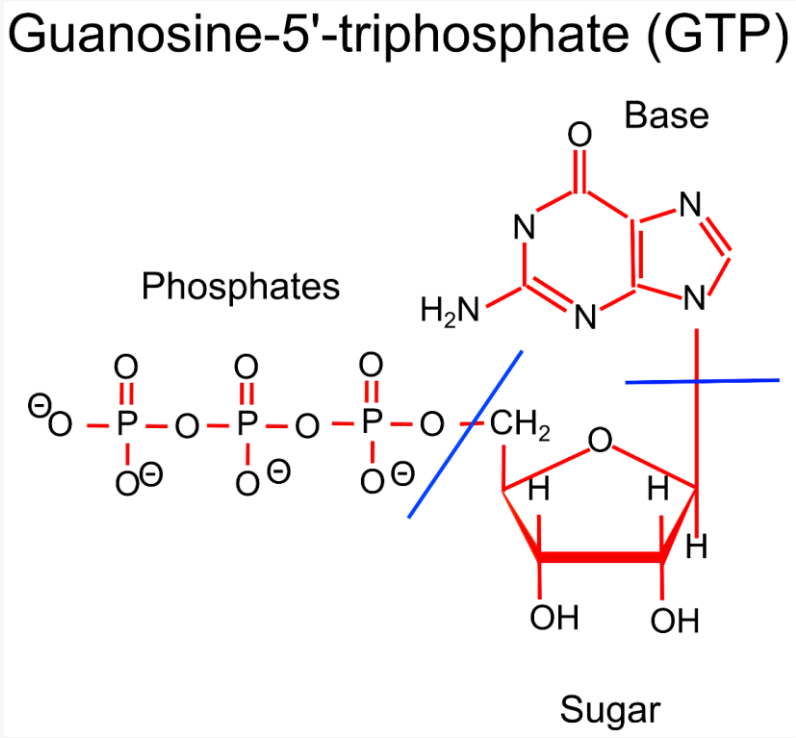
Atenolol (commercial name Tenormin) for treating hypertension (high blood pressure).

Approximately ~30% of drugs are estimated to target this type of receptor.

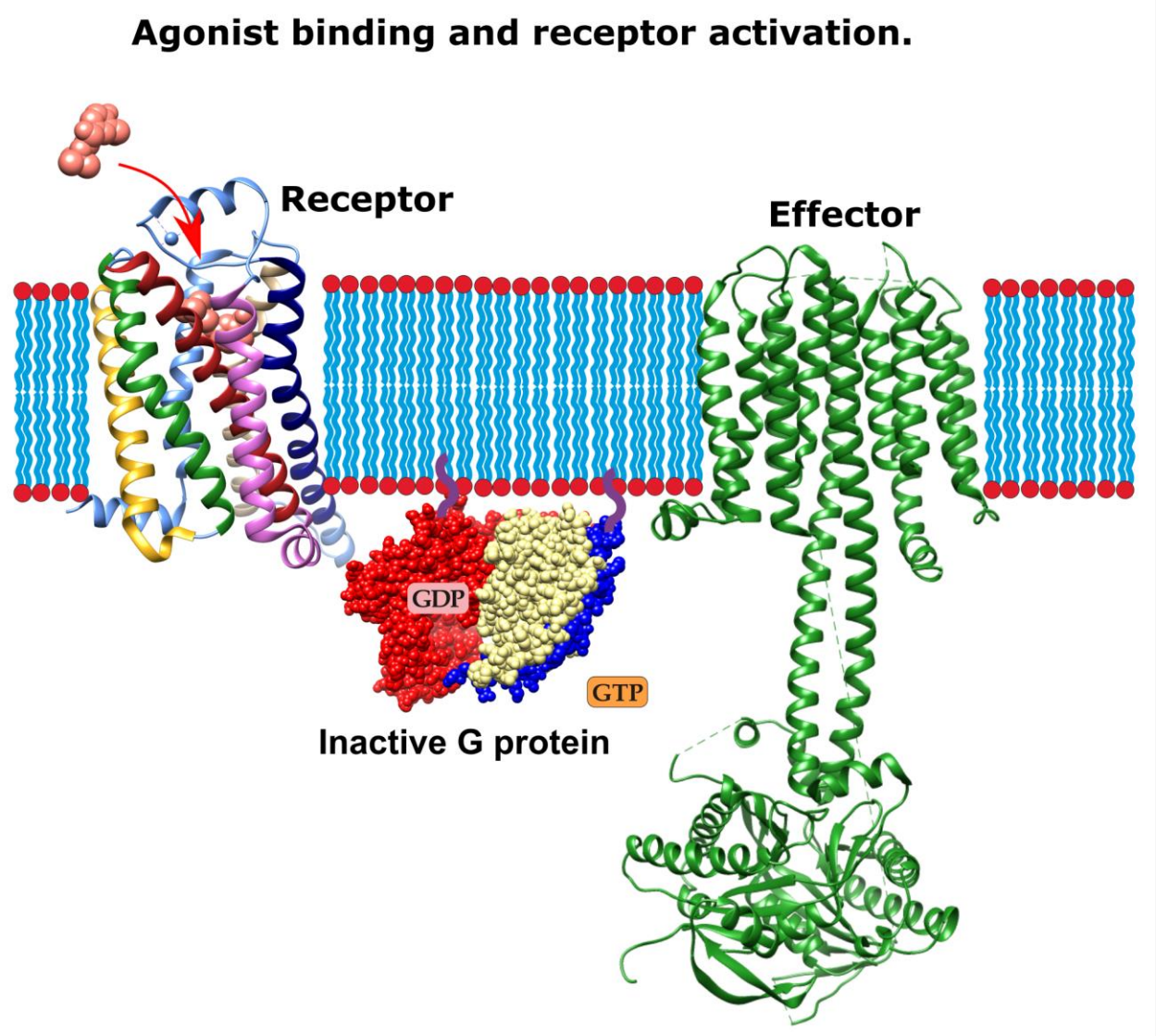
GPCRs interact with heterotrimeric G proteins.

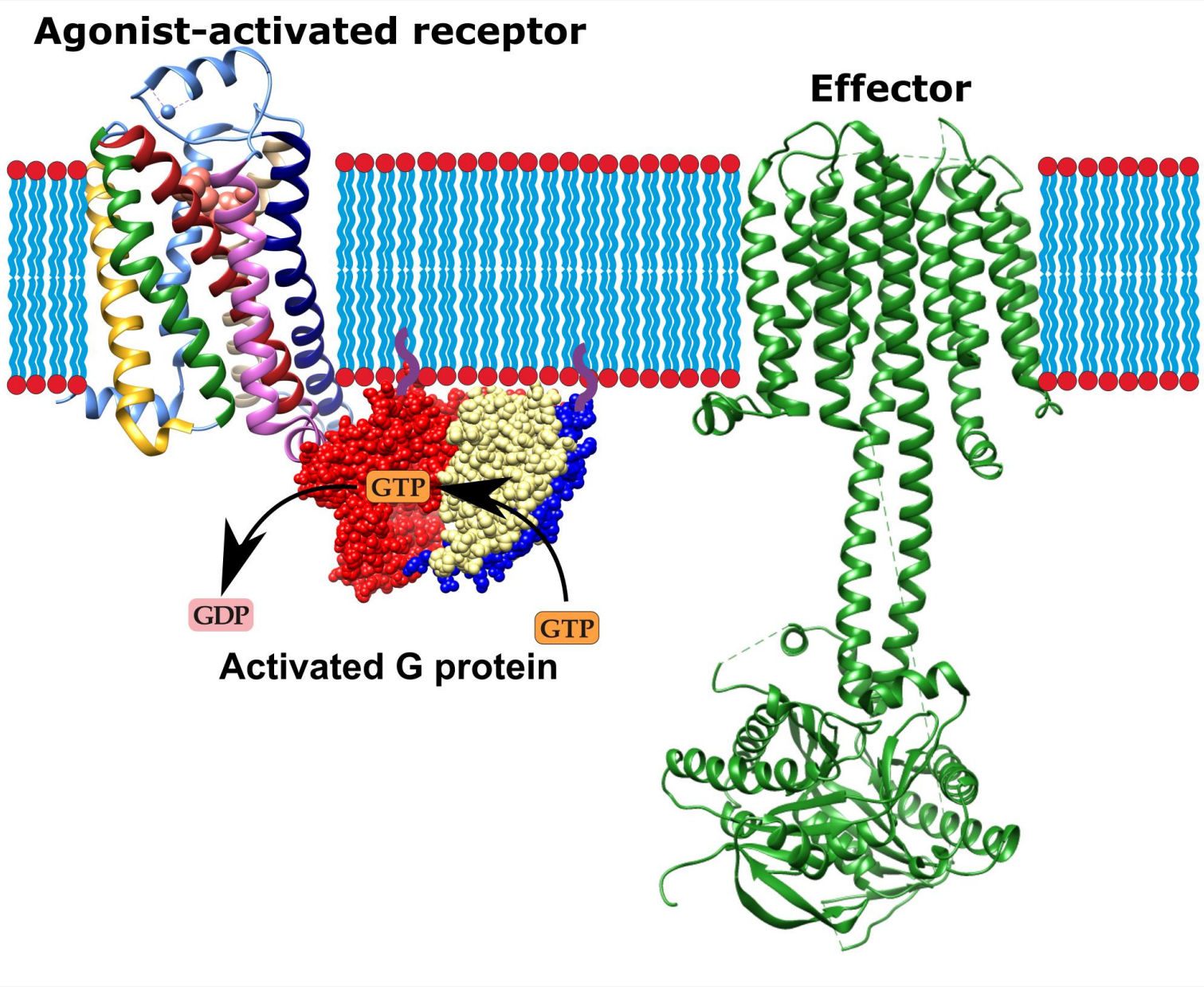
G proteins bind guanine nucleotides, e.g. GTP or GDP.



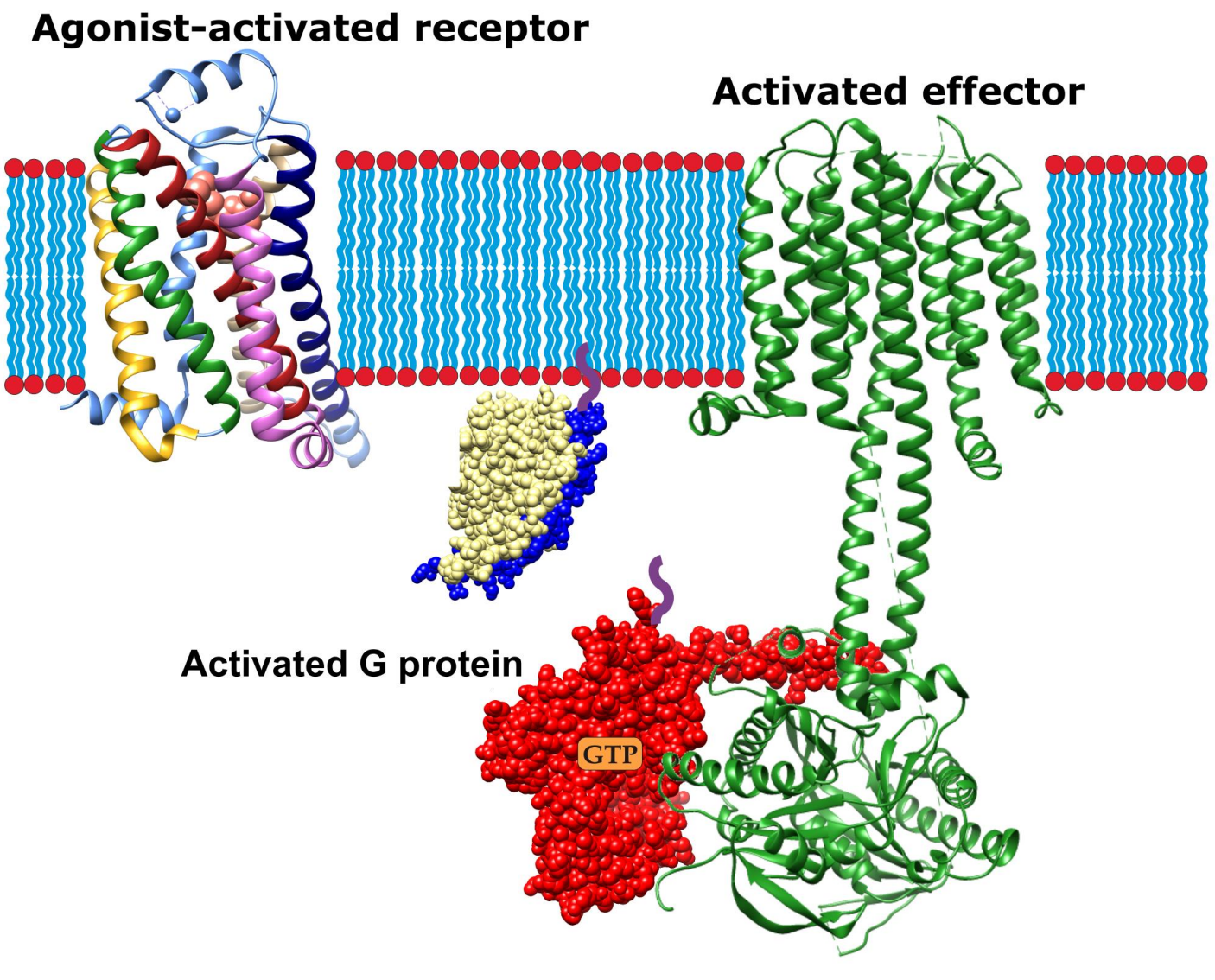


G Protein-**C**oupled **R**eceptor (**GPCR**) cycle.

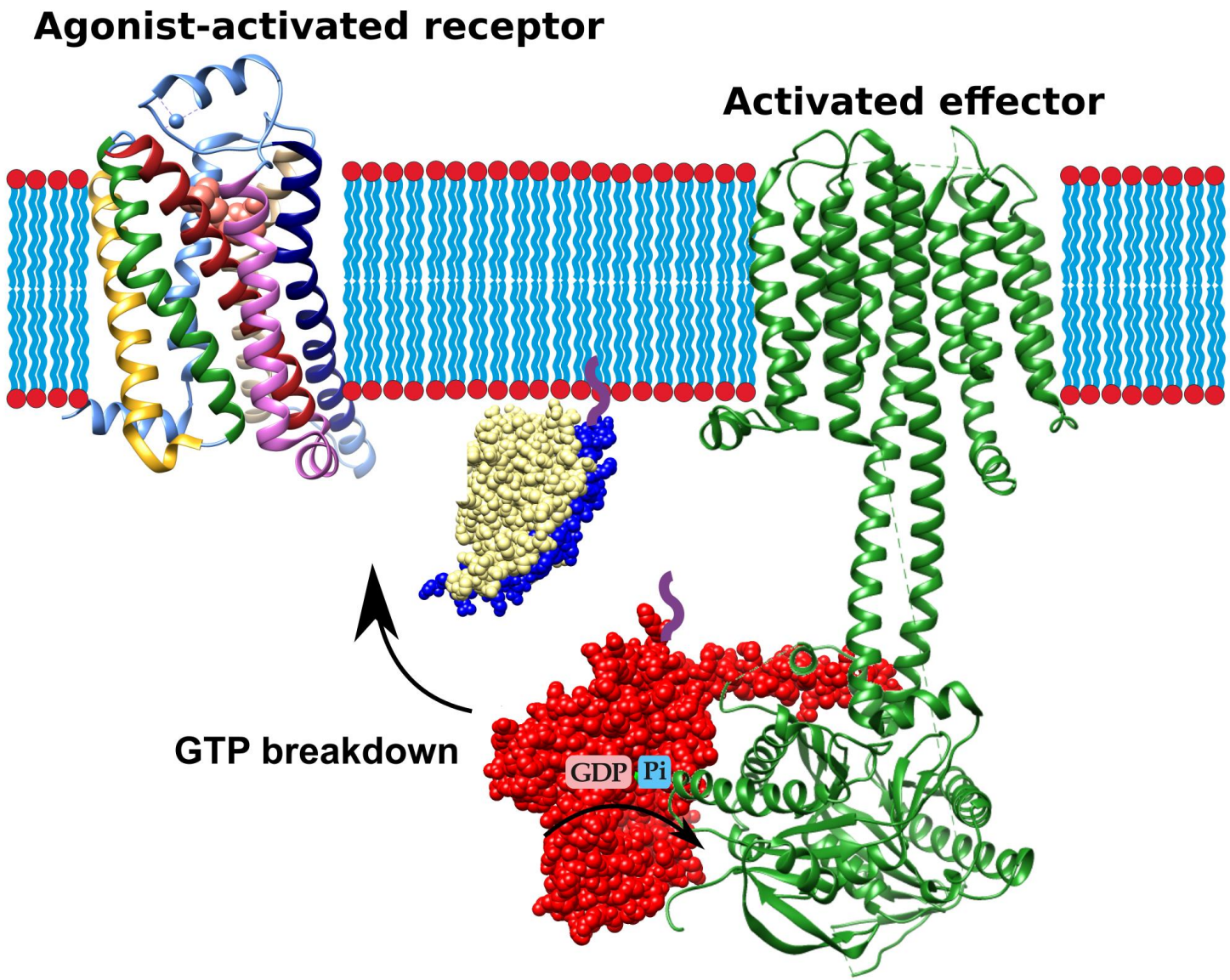




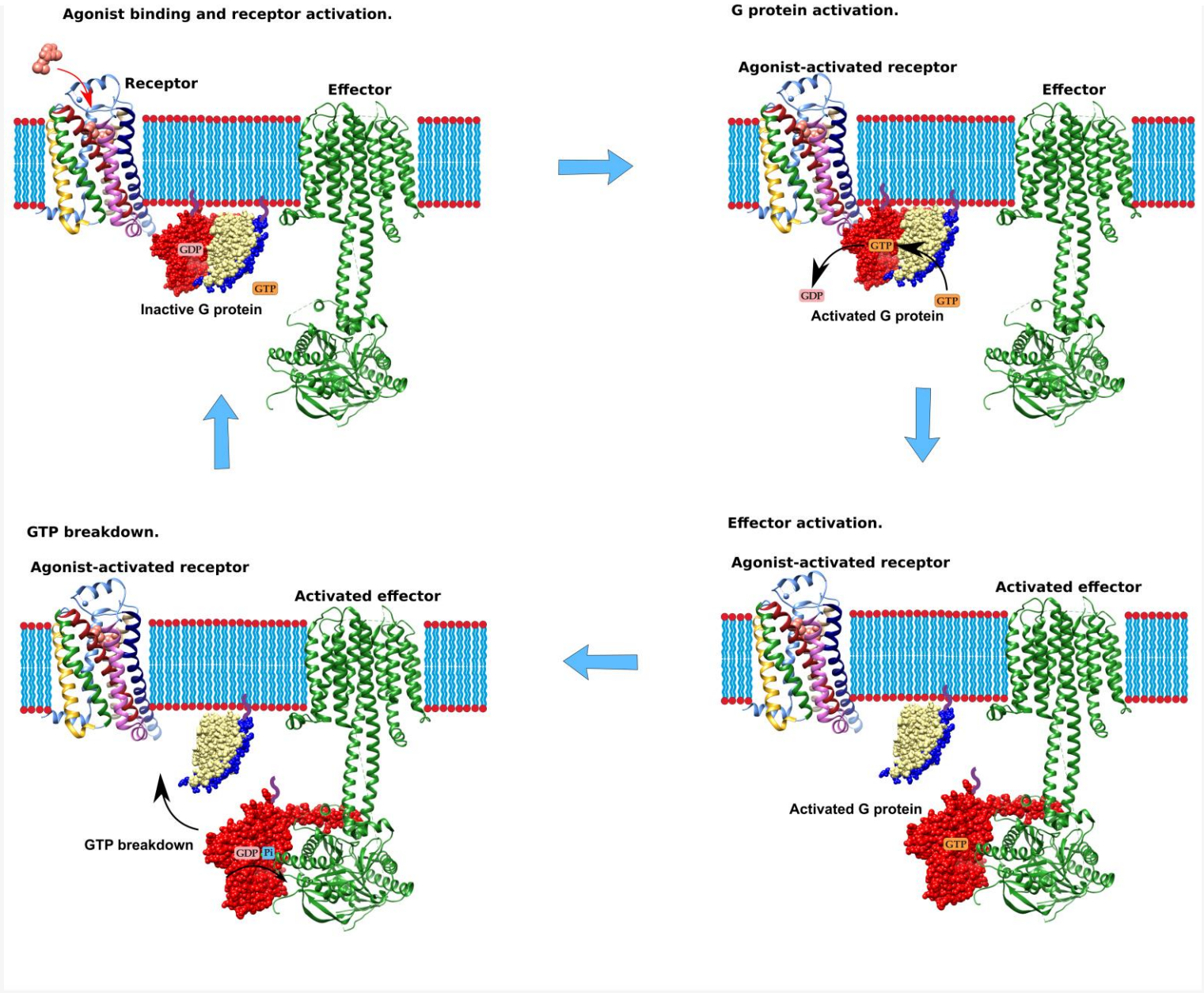
G protein activation



Effector activation.

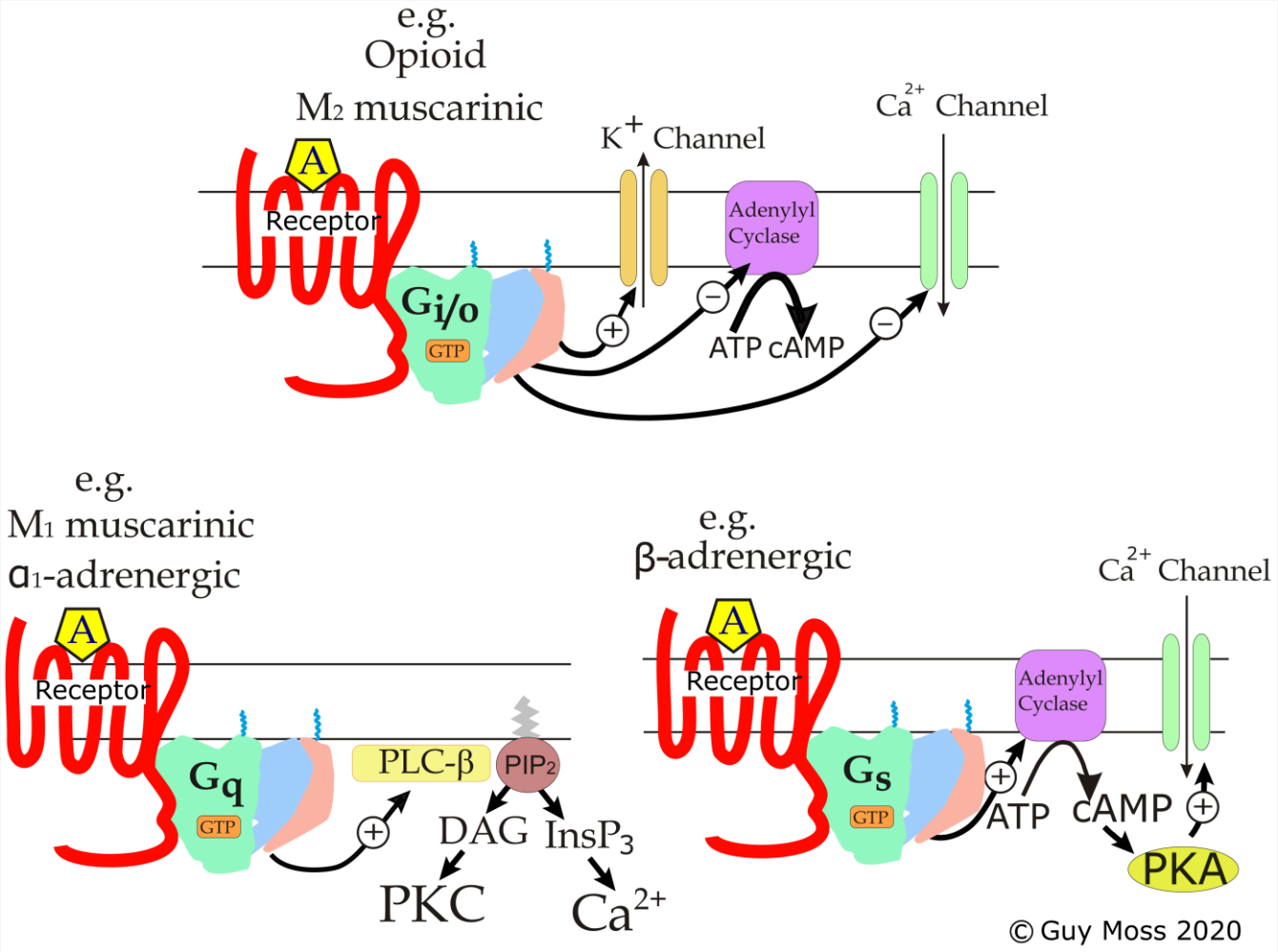


GTP breakdown

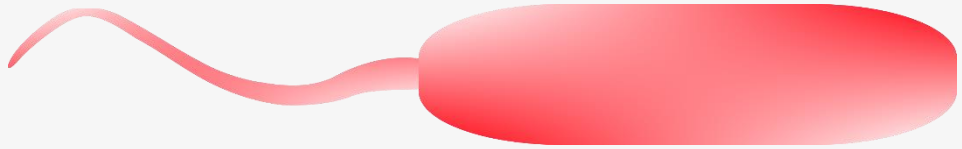


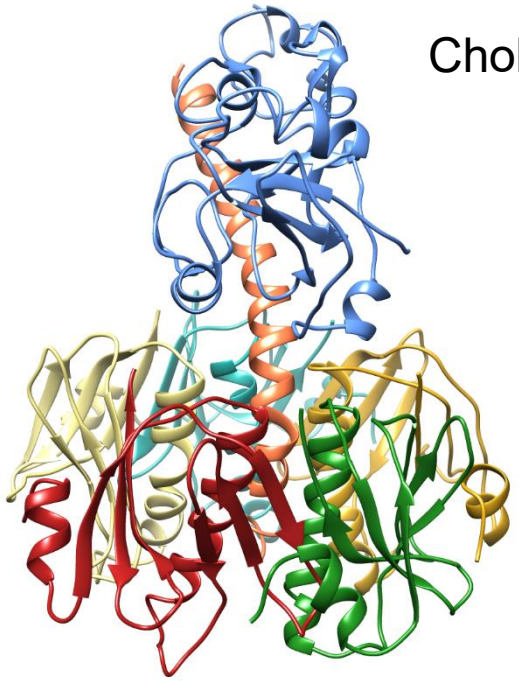
G Protein-**C**oupled **R**eceptor (**GPCR**) cycle.

GPCR coupling to target enzymes and ion channels…



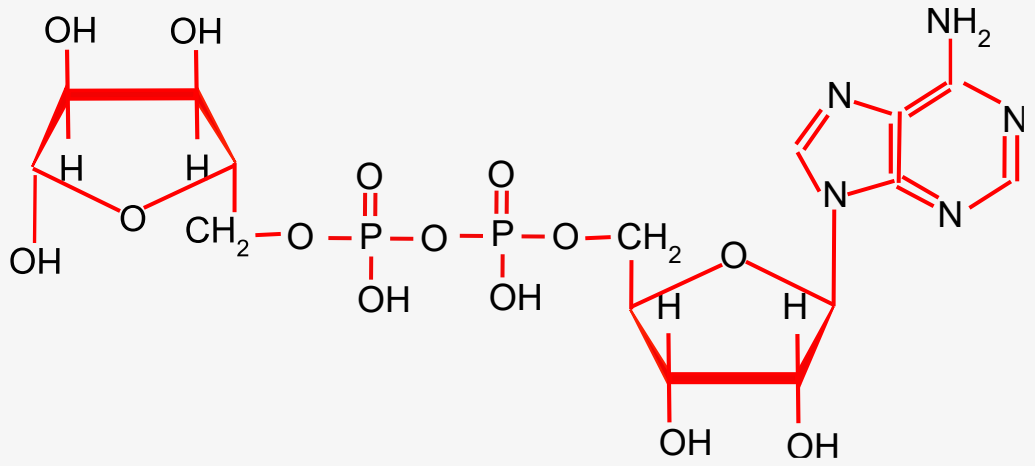
The importance of G-proteins. Vibrio Cholerae…





Cholerae toxin

ADP Ribose



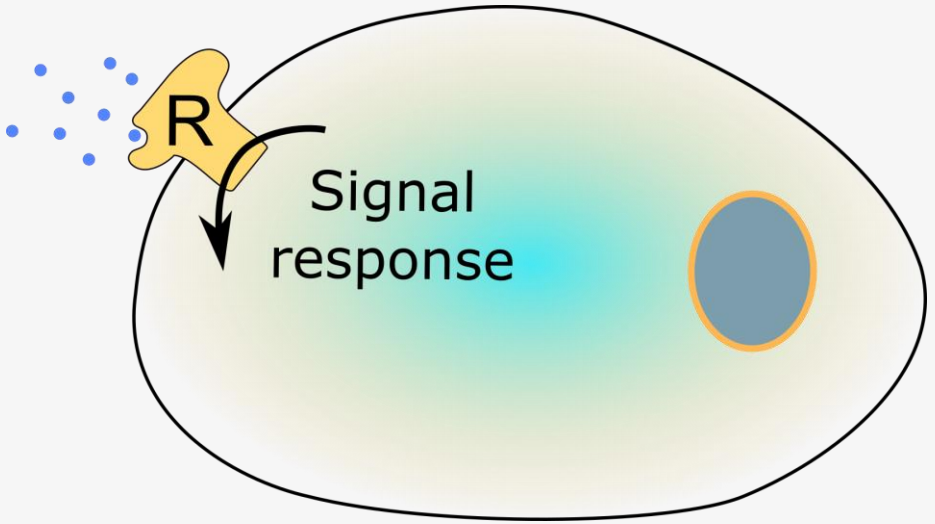
Adrenoceptors, EpiPens and asthma



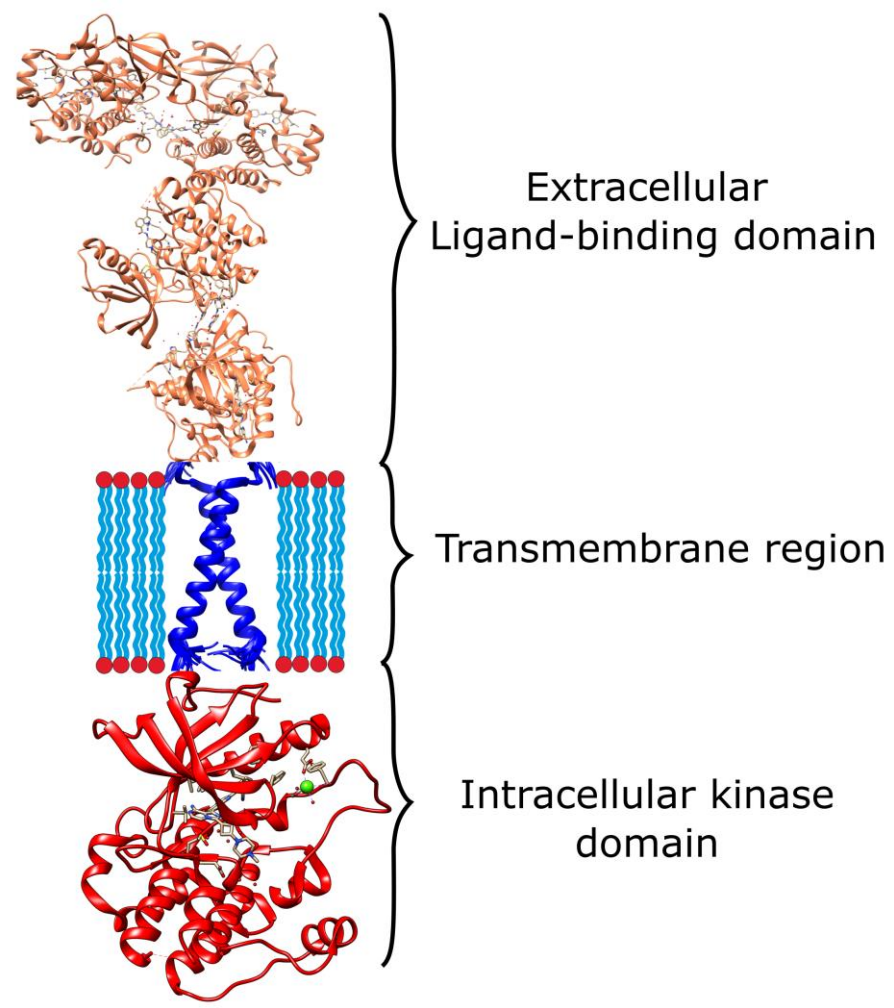


1. G Protein-Coupled Receptors

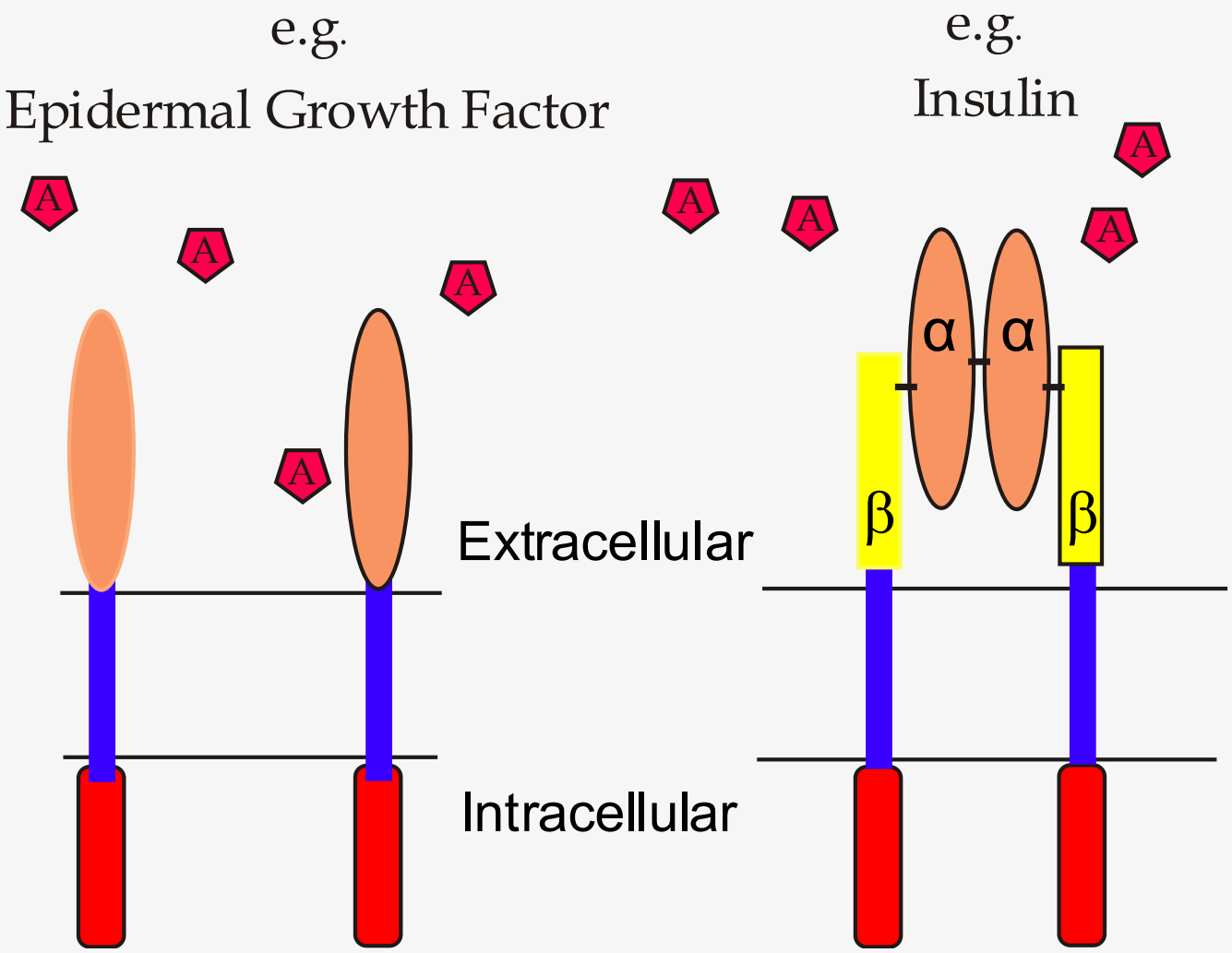
2. Receptor Tyrosine Kinases



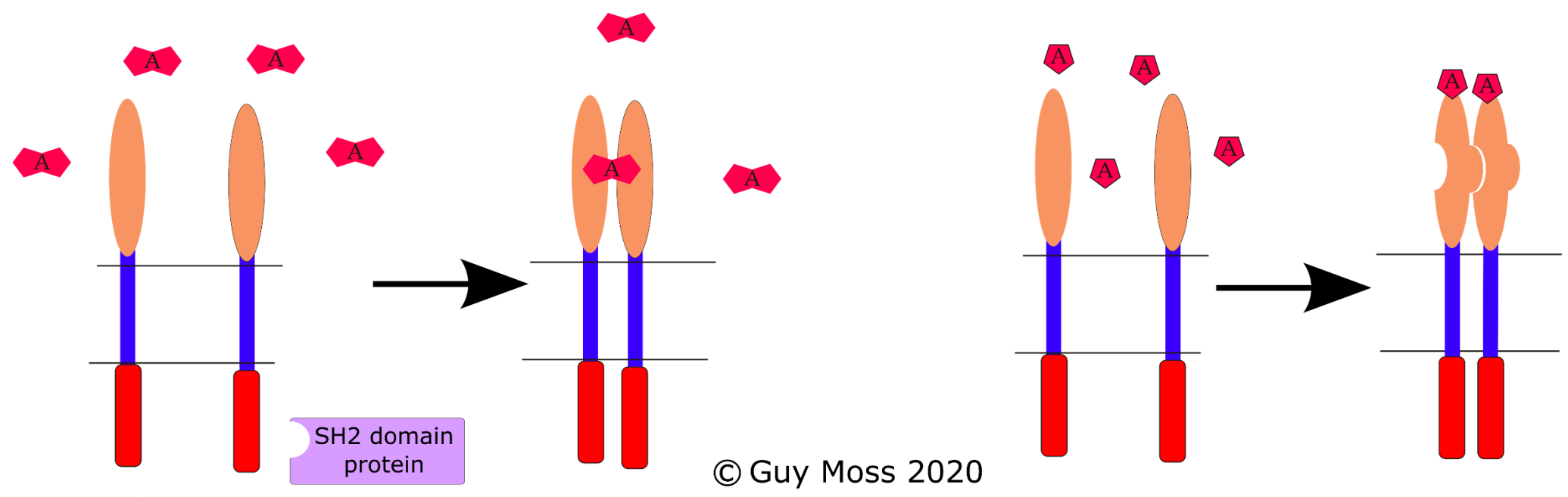
Structure of Receptor Tyrosine Kinases (RTKs)…



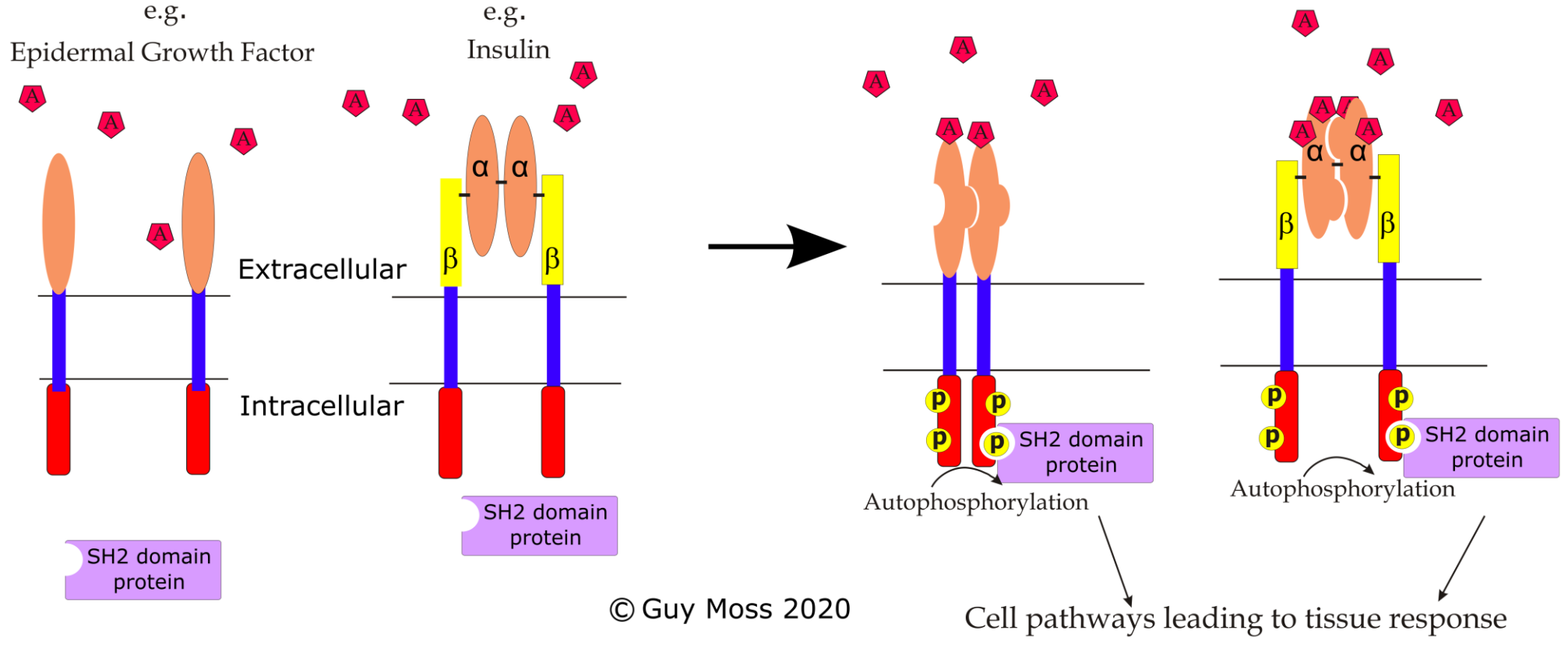
Receptor Tyrosine Kinases.



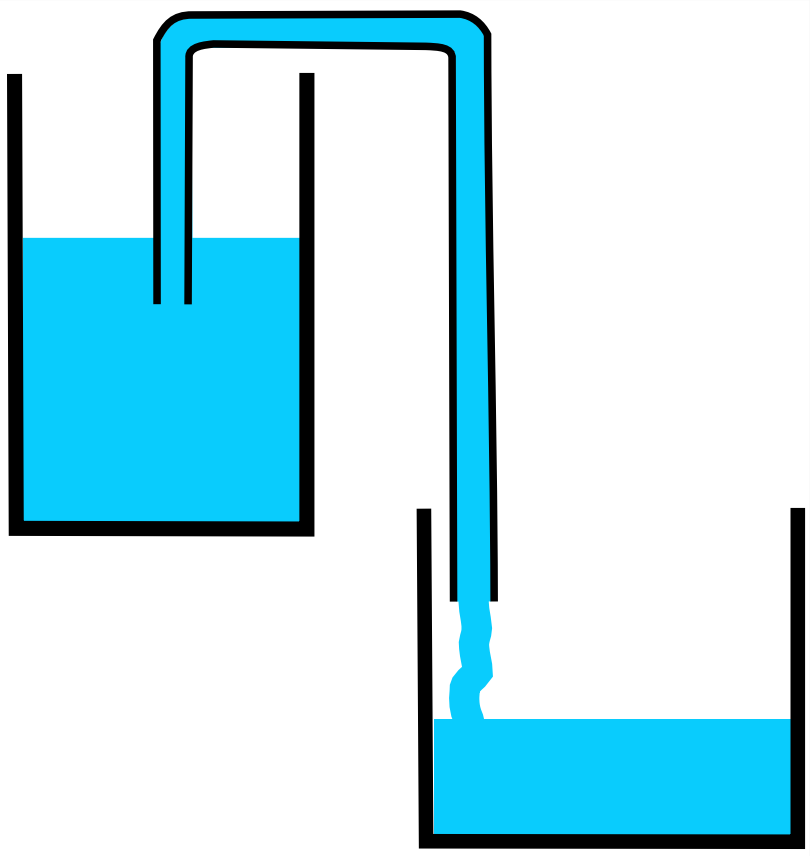
Receptor Tyrosine Kinases.



Receptor Tyrosine Kinases.



Diabetes mellitus…

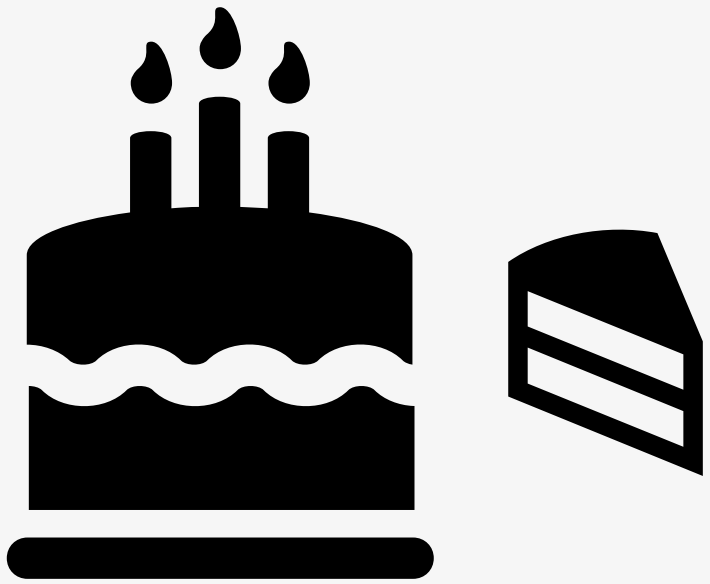


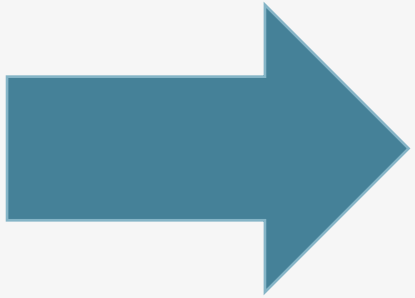
Greek word ‘diabetes’ meaning siphon.



The Latin word for honeyed or sweet is ‘mellitus’ and the honey bee is *Apis mellifera*.

About diabetes mellitus…





Store energy between meals.

Without the necessary supply of energy the body starts to breakdown fats leading to ketoacidosis when the blood becomes more acidic than body tissues. This can be fatal if it continues unchecked.

Mering, Minkowski and Langerhans…

Joseph von Mering and Oskar Minkowski first excised the pancreas of a dog and discovered that it developed the symptoms of diabetes in 1889.

Paul Langerhans discovered the groups of cells that we now call the “islets of Langerhans,” and the noted that their degradation correlated with the onset of diabetes.



Josef, Baron von Mering



Oskar Minkowski



Paul Langerhans

The discovery of insulin…

Fredrick Banting and student Charles Best



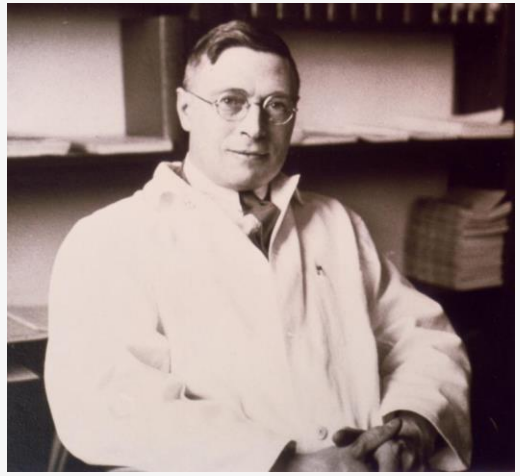
F. G. Banting Papers, Thomas Fisher Rare Book Library, University of Toronto

J.J.R. MacCleod

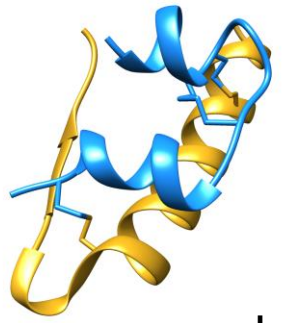


F. G. Banting Papers, Thomas Fisher Rare Book Library. University of Toronto

James Collip



By Scrapbook compiled by Barbara Collip-Wyatt University of Toronto Library



Insulin

On January 11, 1922, Leonard Thompson (14 years of age) became the first person to have his diabetes treated with insulin.

Types of diabetes mellitus…

Type I

- Usually begins in childhood (juvenile onset diabetes).
- Caused by a severe under-production of the hormone insulin,
- Often associated with destruction of the beta cells of the pancreas.
- Also called insulin-dependent diabetes mellitus (IDDM).

Types of diabetes…

Type II

- Usually has a late onset (maturity onset diabetes).
- Characterised by an impaired response to insulin in the body and impaired insulin secretion.
- Weight loss and diet control can be effective treatments.
- Also called non-insulin dependent diabetes mellitus (NIDDM).

Some questions to think about…

1. The body speeds up and slows down the heart via GPCRs. Since both are GPCRs, how can they have such opposite effects?

2. Atenolol mentioned at the start of this lecture is used to prevent blood pressure rising too high. A raised blood pressure occurs when you have a 'fight or flight' response. So how might atenolol work?